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Influencing mechanisms of live streamer's language strategies on product sales

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ABSTRACT

Live-streaming e-commerce has experienced rapid growth in recent years. Drawing on Hovland's persuasion theory and Aristotle's rhetorical theory, this study categorizes streamers' persuasive linguistic strategies into six types: rewards and exaggeration, rewards and logic, rewards and emotion, credibility and exaggeration, credibility and logic, and credibility and emotion. Through text analysis and statistical modeling, the results show that "rewards and exaggeration," and "credibility and exaggeration," positively influence sales, while "credibility and logic" has a negative impact on sales. The study further finds that customer engagement mediates the relationship between most linguistic strategies and sales, while electronic word-of-mouth (eWOM) amplifies the influence of customer engagement on sales. This research enriches theoretical insights into persuasive language use in live-streaming e-commerce and provides practical recommendations for streamers aiming to optimize engagement and boost sales performance.

1. Introduction

Live streaming e-commerce is an emerging business model that provides consumers with shopping services through real-time images, sounds, and rich interactions (Mao et al., 2022; Li et al., 2024). On June 9th, 2022, streamer Dong Yuhui gained popularity for his "charm of languages," which increased his live stream sales by more than 16 times in less than two weeks (Vivian, 2022). As the number of live streamers has grown and the competition among different e-commerce platforms has intensified, customer retention has become increasingly challenging (Yan et al., 2023). Streamers' language plays a crucial role in live streaming by enhancing consumers engagement and purchase intentions, (Yao et al., 2024; Luo et al., 2021). Compared with deciding what you say, it is more important to decide how you say (Wu et al., 2017). Language persuasion and language content are distinct; language content represents what is communicated, while language persuasion reflects how language is used to influence or align others with the speaker's goals (Luo et al., 2021; Parhankangas and Renko, 2017). For live streamers, linguistic style is essential, as it allows them to captivate and retain viewer interest, even in unscripted moments (Liu et al., 2023a). However, few studies have systematically studied persuasiveness in the form of text, especially persuasive language and its impact on consumer behavior (Luo et al., 2021). Meanwhile, brand suppliers and streamers face the challenge of using language to expand and retain their consumer base in live stream shopping, catering to consumer psychology, designing effective live stream shopping content, and boosting sales.

Live Streamer plays a significant role in driving product sales within live stream shopping. Previous studies have extensively investigated how streamers characteristics, including nonverbal communication (Liu and Zhao, 2024), streamers' similarity and professionalism (Li et al., 2024), genders (Yang et al., 2023), beauty, expertise, humor, and passion (Guo et al., 2022), affect consumers' cognition, emotion, and behavior intention. However, studies on the streamer's persuasive language are still in its infancy. For example, Li et al. (2023) identified four dimensions of persuasive language style: appeals to emotion, logic, credibility, and social that positively influence tourists' willingness to travel in live streamed travel content. Hao & Huang (2024) investigated the impact of scarcity-driven persuasive messages within the context of live-streaming e-commerce. The persuasiveness of a streamer's message can be shaped by the various linguistic styles (Holtgraves, 2001). However, the effects of persuasive language in live streaming e

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commerce remains insufficiently understood. Addressing this gap, our study focuses on the linguistic combinations streamers use, based on persuasion theory and rhetorical theory, to create persuasive and engaging interactions, aiming to establish a comprehensive framework that guides the effective use of language to maximize customer engagement and product sales in live-stream shopping. To put it differently, brand suppliers and live streamers, whether already employing or aiming to improve their language strategies for a competitive edge, could enhance customer engagement and boost product sales by understanding persuasive language strategies. Consequently, it is essential to further investigate the impact of persuasive language in live-stream commerce, particularly focusing on effective combinations of linguistic persuasion techniques.

Our study draws on two foundational theories: Hovland's Persuasion Model and Aristotle's Rhetorical Theory, each of which provides new perspective on the mechanisms of persuasion in live-stream e commerce contexts. Hovland's Persuasion Model posits that the effectiveness of a message depends on four elements—persuader, persuasive information, persuasion context, and the audience response—that together influence the cognitive and attitudinal transformations of the audience (Hovland and Weiss, 1951; Hovland and Weiss, 1951). Hovland persuasion model suggests that the persuader, as the primary factor in the process of persuasion, must have credibility, which is essential for effective persuasion. Persuasion process begins with the disseminator, with their credibility serving as the basis for persuasion (Li et al., 2023). Therefore, in the context of live-stream e-commerce, streamers serve as both persuaders and sources of persuasive information, shaping consumer attitudes and behaviors through a combination of persuasive information (language strategy), tailored towards desired outcomes. Besides, the streamer acts as an authoritative figure, and their credibility language influences purchase intention (Luo et al., 2021). Hovland's model thus provides a foundational structure for examining how these language strategies drive engagement and sales (Li et al., 2023).

In addition to providing product information, streamers often intersperse rewards, such as giving out gift bags, taking screenshots, and awarding prizes, to retain customers and keep them watching and engaging in the live stream. This dynamic and real-time interaction enables streamer to continuously adjust their language (Xue et al., 2020), creating a unique opportunity to study how specific linguistic strategies impact customer engagement and sales. Reward and exaggeration, as part of the content style in Hovland's persuasion model, can influence the attitude of the information receiver towards the product, reflecting two primary approaches used by streamers (Luo et al., 2021). The reward category leverages emotional appeals by focusing on tangible benefits such as discounts or exclusive deals, which can create excitement and a sense of urgency. This reward-based language is particularly effective in live-stream settings where time-limited offers encourage impulsive buying behavior. Employing various linguistic persuasion techniques to convey the same content can lead to varying levels of persuasive effectiveness (Chen et al., 2024). Streamers frequently employ exaggerated language during live broadcasts to spark consumer interest in the product (Luo et al., 2021). Building on Hovland's persuasion model, Luo et al. (2021) studied the influence of streamer's linguistic styles, such as personality and logic, on the sales of live stream products. However, their work did not classify these linguistic styles into combinations with rhetorical expressions and systematically analyze their effects.

Aristotle defined "rhetoric" as the ability to identify persuasive techniques in any situation, emphasizing on how mood and morality influence audiences and assessing persuasion through three key elements: credibility, logic, and emotion (Aristotle, 1954; Baumrin, 1986). Aristotle's rhetorical theory has been used in various research fields, including live streaming e commerce context, where it has been shown to improve customer behavioral intention (Chu et al., 2014; Tirdatov, 2014). Research has confirmed the applicability of his rhetorical theory in live streaming e commerce context (Li et al., 2023; Luo et al., 2021). While previous studies have confirmed the applicability of Aristotle's theory in live-streaming contexts, none have systematically examined how combinations of rhetorical elements influence consumer behavior. Although Li et al. (2023) applied Aristotle's rhetorical theory to examine streamers' persuasive language, their study lacked empirical analysis, relying instead on conceptual exploration. Our study addresses this gap by leveraging objective data and employing econometric modeling to analyze the effects of different linguistic combinations on customer engagement and product sales. To address this gap, our study integrates Aristotle's Rhetorical Theory with Hovland's model to classify persuasive language into two categories: reward-based and credibility-based approaches. The credibility category comprises language that emphasizes trustworthiness, product quality, and reliability—key factors for building confidence in both the product and the streamer. Streamer in live streaming often combine various rhetorical techniques and speech strategies to achieve persuasive goals. To explore the impact of different styles of linguistic persuasion on consumer behaviour, we combining Aristotle's Rhetorical Theory and Hovland's model, we classified persuasive language used in reward scenarios into three categories: rewards and exaggeration, rewards and logic, and rewards and emotion; similarly, the combinations of persuasive language for product introductions were categorized into combinations of credibility and exaggeration, credibility and logic, and credibility and emotion. This leads to our first research question of this study: How do different persuasive language combinations used by live streamers impact customer engagement and product sales?

Additionally, Aristotle's distinction between artistic and non-artistic persuasion underscores the importance of electronic word-of-mouth (eWOM) as a non-artistic factor (Baumrin, 1986). Unlike artistic persuasion, which is crafted by the persuader, non-artistic persuasion involves external sources of credibility, such as customer ratings. In live-stream shopping, eWOM plays a critical role in enhancing consumers' perception, engagement and guide their purchase behaviors (Haq et al., 2024; Wang and McCarthy, 2021; Mehra, 2023). By showcasing positive reviews or customer feedback during a live stream, streamers can use eWOM to amplify the impact of their persuasive language, thus encouraging greater engagement and conversions. However, previous studies have largely treated eWOM as a control variable (e.g. Ma et al., 2024). This study, in contrast, examines eWOM as a non-artistic persuasion moderating factor, considering its potential to influence the relationship between customer engagement and product sales. Therefore, this study introduces electronic eWOM as a non-artistic persuasion element, leading to our second research question: How does non-artistic element eWOM influence the relationship between customer engagement and product sales?

Key factors influencing attitude changes include the persuader, the persuasive information, and the persuasion context, serve as external stimuli. Recipient of persuasive messages is not merely passive; rather, their attitudes shift based on personal predispositions and systematic responses to these external influences (Hovland, 1959). Therefore, linguistic persuasion, is a method for guiding attitudes and behaviors toward a persuader's intended outcome. In marketing, research on persuasion theory has gained traction. For instance, Zboja et al. (2021) investigated the connection between persuasive knowledge and sales pressure, and Isaac and Grayson (2017) found that persuasive knowledge acquisition fosters trust in information sources. Yang (2015) studied how individual traits influence purchasing behavior in e-commerce within the framework of persuasion theory. Research has demonstrated that streamers' language styles and choice of topics influence consumer engagement behaviors, which subsequently affect attitudes and sales outcomes (Ma et al., 2024; Luo et al., 2024a). Although prior studies have analyzed the direct effects of rational, emotional, credible, and reward-based language on purchase intentions and customer engagement (Dolan et al., 2019; Luo et al., 2021), the mediating role of persuasive linguistic combinations in fostering customer engagement and enhancing audience connections remains

insufficiently explored. This leads to the study's third research question: What is the mediating role of customer engagement in the relationship between persuasive language combinations and product sales?

In the pursuit of answers to these questions, natural language processing (NLP) and deep learning technology was used to analyze streamer speech, and the paradigms and methods of empirical research were used to explore the influence of streamer's persuasive linguistic combinations on consumer engagement and product sales.

In summary, grounded in Hovland's persuasion model and Aristotle's rhetorical theory, this study demonstrates that streamer's persuasive language combinations influence product sales, with customer engagement plays as an intermediary role. Our study provides three substantial theoretical contributions. First, this study enriches the understanding of the efficacy of streamer verbal persuasive in customer engagement and product sales in live stream shopping and contributes to the research on both customer engagement and product sales. Second, this study expands Aristotle's rhetorical theory into live stream shopping scenarios by investigating the combination of persuasive language with Hovland persuasive model. Furthermore, it validates the positive moderating effect of non-artistic persuasion, our study delves into the moderating impact of eWOM, specifically focusing on non-artistic persuasion factors in live stream shopping on product sales. This distinctive approach makes a novel contribution to the body of research pertaining to live streamer's language strategies. The findings reveal the new strategies of streamer persuasive linguistic for digital marketers. In the future, live streaming platforms, streamers, and suppliers could apply these findings to refine persuasive language strategies and advance live-stream shopping development.

The remainder of this study is organized as follows. First, we provide a comprehensive review of the relevant literature on persuasive language in digital marketing, customer engagement, and electronic word of mouth (eWOM). Next, we present the hypotheses and the research model in Section 3. Following this, we describe the research setting and outline the methodology used for data collection. In section 5, we present the results of both statistical and econometric analyses. Finally, we summarize the key findings, highlight the theoretical and practical contributions, discuss the study's limitations, and propose potential directions for future research.

2. Literature review

2.1. Persuasion language in digital marketing

Since antiquity, researchers have sought to identify rhetorical techniques that enhance a speaker's persuasive effectiveness (e.g., Aristotle, 350 BC). The study of persuasion dates back to World War I and II, when governments and organizations recognized the significant influence of communication in shaping public opinion and behavior (LaBrecque, 2008). Language cues and communication techniques serve as effective tools for persuasion (Hosman, 2002). This acknowledgment led scholars to explore how persuasive language affects attitudes and beliefs, contributing to the development of the field known as "persuasion and attitudes research" (Joyal-Desmarais et al., 2022). It also spurred the application of persuasive techniques in modern digital advertising, including online applications, digital ads, recommendation systems, chatbots, and e-commerce platforms (Braca and Dondio, 2023). Persuasion is vital in marketing, allowing marketers to engage effectively with consumers and influence their decisions to achieve business goals (Braca and Dondio, 2023). The primary objective is to impress target audiences or attract potential customers, driving desired market changes and boosting product sales (Çam, 2019).

Persuasion is characterized by the manner and presentation that enhance the power of words and ideas, aiming to influence decision-making and encourage compliance (Asemah and Edegoh, 2013; Crittenden and Crittenden, 2023; Rimer and Kreuter, 2006; Shahid et al., 2017). Persuasion involves using targeted strategies to influence others' perceptions and actions, guiding them to accept certain views and achieve a specific goal (Amos et al., 2021). It directs the attitudes and behaviors of the recipient according to the persuader's intent (Chen et al., 2024). Linguistic style plays a vital role in persuasive communication, enabling the conveyance of subtle meanings and nuances within messages (Braca and Dondio, 2023). When it comes to persuasion style, it talks about what strategies are practiced in persuasion (Burgers and de Graaf, 2013).

The persuasive context refers to the setting in which information is conveyed, influencing the audience's cognitive interpretation and emotional reaction to it. (Hovland and Weiss, 1951; Leong et al., 2021). Persuasive communication in digital marketing encompasses various linguistic styles that influence the effectiveness and persuasiveness of a message. These styles include discourse modes, delivery, diction, tone, genre, cues, and personal or professional nuances, which collectively influence how a message is perceived and acted upon (Górska, 2019; Leong et al., 2021; Liu et al., 2023a; Zhou and Huang, 2024). Holtgraves (2001) argued that linguistic styles could modify the process and outcomes of a message's main assertion, thereby affecting the success of persuasive communication. The response to a persuasive claim can be influenced by various factors, such as the communicator's expertise, cultural background, and the presentation's length and content (Lasswell, 1948).

The rapid evolution of live-streaming e-commerce has transformed the way consumers engage with digital shopping platforms, creating an interactive space where real-time persuasion and consumer interaction play a critical role (Feng et al., 2024). The immediacy and interactivity of live streaming provide unique opportunities and challenges for applying persuasive strategies (Li et al., 2023). Compared with traditional e-commerce, live streaming e-commerce can provide a more vivid audio–visual experience and establish an interpersonal relationship between the streamers and users (Wang et al., 2020). Effective persuasive communication in live streaming influences consumer attitudes, perceptions, and purchasing decisions (Li et al., 2023). Linguistic style is particularly significant for live streamers, as it helps maintain viewer interest through strategic language use, which is closely associated with sales performance (Liu et al., 2023a). While research has begun to explore various linguistic style such as task oriented and social oriented style (Yao et al., 2024; Wu et al., 2023; Xu et al., 2022), sensory language (Hu and Ma, 2023) and language appeals (Ma et al., 2023a), as well as the perceived linguistic persuasiveness of live streamers (Chen et al., 2024), the role of comprehensive persuasive language strategies remains underexplored. The specific use of vocabulary and the method of presenting information can elicit diverse reactions from audiences, influencing their perceptions, intentions to purchase, and actions (Searle, 1969).

Previous studies have examined persuasive strategies from the perspectives of psychology, philosophy and communication (Braca and Dondio, 2023), focusing on elements such as argument quality (Keskin et al., 2017), rhetorical techniques (Stiff and Mongeau, 2016), language power (Gadzhiyeva and Sager, 2017), two-sided messages (Sheng et al., 2022). As summarized in Table 1, while research has explored persuasive linguistic styles, most studies have concentrated on specific appeals, such as emotional or rational appeals. However, previous studies have not fully integrated or empirically tested the comprehensive persuasive impacts of a combination of linguistic appeals within live-streaming commerce. Although few studies have examined the impact of persuasive language on live streamers' persuasiveness (e.g. Luo et al., 2021; Li et al., 2023; Hao and Huang, 2024), there remains a significant gap in understanding how specific linguistic strategies, such as credibility and reward-based language combine with rhetorical elements, directly affect consumer behavior and sales performance in live streaming e-commerce. Further theoretical examination of live streamers' persuasive linguistic style as a critical verbal cue in this context is needed. Besides, most existing data come from online experiments or surveys, with few studies using objective data from real-time live-streaming platforms.

Table 1
Summary of existing studies on language research in online commerce.

Reference	Context	Methods	Independent variable(s)	Mediating variable	Dependent variable	Moderating variable	Research focus	Theory
Online Commerce								
Keskin et al. (2017)	Social media marketing	Qualitative	Argument quality, Post popularity, Post attractiveness	Sympathy	Empathy	–	Persuasive message	Elaboration likelihood model theory
Gadzhiyeva and Sager (2017)	Sales person	Questionnaires	Language Power, Nonverbal Immediacy, Biological Sex	Perceived speaker power	Extent of Persuasion	–	Language power	–
Wu et al. (2017)	Online review	Online scenario-based experiments	Figurative vs. Literal language	Language typicality and perceived reviewer expertise	Attitude, reservation intention	Reviewer's expertise level	Language style	Language Expectancy Theory
Sheng et al. (2022)	vloggers	Questionnaires	Credibility of two-sided messages persuasion, Appeal of emotional persuasion	Perceived product value, Para-social interaction	Consumers' purchase intention	–	Persuasive strategies	Aristotle's persuasion theory
Live streaming e-commerce								
Luo et al. (2021)	Live streamer	Latent Dirichlet Allocation (LDA) model	Host's perspective: Personality, Reward Rhetorical perspective: Emotion, Logic, Exaggeration	–	Sales	Product category	Persuasive linguistic styles	Hovland's persuasion theory and Grounded theory
Li et al. (2023)	Travel live streamer	In-depth interview	Appeals to Emotion, Logic, Credibility, Social, Speech cues	–	Travel intention	–	Persuasive linguistic styles	Hovland's persuasion theory and Aristotle's rhetorical theory
Hu and Ma (2023)	Virtual live streamer	Online scenario-based experiments, Focus group	Sensory language vs non sensory language	Language expectancy violation	Purchase intention	Human-backed or AI-backed virtual streamers	Sensory language	Language expectancy theory
Ma et al. (2024)	Live streamer	Questionnaires	Rational Appeal, Emotional Appeal	self-referencing, self-brand congruity	Purchase intention	–	Language appeals	Stimulus-Organism-Response (SOR) theory
Luo et al. (2024b)a	Live streamer	LDA topic modeling	Informative topic, Affective topic	–	Users' Behavioral Engagement, Users' Emotional Engagement, Users' Relational Engagement	–	Streamers' topic types	Social support theory
Chen et al. (2024)	Live streamer	Questionnaires	Resposiveness, Entertainment, Personalization, Emotional contagion, Expertise, Logic, Morality	Affective distance, Affective trust, Cognitive trust		–	Persuasion ability & Interaction ability	interaction theory, affective distance theory, trust theory, and Aristotle's rhetorical appeals
Hao and Huang (2024)	Live streamer	Scenario-based experiment	Quantity-based scarcity induced appeals	Arousal	Impulsive buying	Timed-based scarcity induced appeals	Persuasive effect of different scarcity messages	Competitive arousal model
Yao et al. (2024)	Virtual live streamer	Online experiments, Lab experiment, Focus group	(linguistic style: social-oriented vs. task-oriented) × (product type: experience vs. search)	Perceived warmth, Perceived competence	Purchase intention	Virtual streamer types	Linguistic style	The stereotype content model
Ma et al. (2024)	Live streamer	Econometric model	Representative act, Expressive act, Commissive act, Directive act	Customer engagement	Purchase behaviour	–	Speech act	Speech act theory

(*continued on next page*)

Table 1 (*continued*)

Reference	Context	Methods	Independent variable(s)	Mediating variable	Dependent variable	Moderating variable	Research focus	Theory
This study	**Live streamer**	**Econometric model**	**Linguistic persuasive combinations:** Reward and exaggeration, Reward and logic, Reward and emotion, Credibility and exaggeration, Credibility and logic, Credibility and emotion	**Customer engagement**	**Product sales**	**eWOM**	**Persuasive linguistic styles**	**Hovland's persuasion theory and Aristotle's rhetorical theory**

Additionally, the combined role of various persuasive linguistic strategies has rarely been explored in detail. Recognizing the effectiveness of rhetorical persuasive language has the benefits of effective way for persuading individuals, this study analyzes live streamer speech data to examine the influence in real-time user engagement on product sales.

The current study builds upon this body of literature by specifically focusing on how different combinations of persuasive language (e.g., credibility and logic, reward and emotion) affect consumer engagement and sales in live-streaming e-commerce. Unlike previous research that has emphasized specific linguistic style/rhetorical or the attributes of streamers, this study uniquely integrates verbal persuasive strategies with audience response and the moderating effect of eWOM. By applying Hovland's Persuasion Model and Aristotle's Rhetorical Theory, this provides a dual-theory framework that explores how artistic (language strategies) and non-artistic (eWOM) elements interact to influence viewer behavior.

2.2. *Customer engagement*

Customer engagement has emerged as a critical determinant of business success in today's digital environment (Ma et al., 2023b). However, customer engagement has been defined and operationalized in diverse ways across the literature. To provide a coherent foundation for this study, we first review and synthesize key definitions of customer engagement and then explain their relevance for our research context.

Vivek (2009) defines customer engagement as a behavioral concept, that is, the extent to which customers actively participate in and remain connected with an organization's products or services. Building on this, Brodie et al. (2011) identify three dimensions: cognitive, emotional, and behavioral. Cognitive engagement refers to the consumer's level of mental focus and interest in a brand, emotional engagement represents the affective investment in the brand, and behavioral engagement captures the time, effort, and energy consumers devote to brand interactions (Hollebeek et al., 2014). Islam and Rahman (2016) further describe customer engagement as the willingness of consumers to interact with and participate in a particular entity, such as a website, organization, brand, or community. Customer engagement in commercial activities manifests in diverse forms, encompassing both monetary behaviors (e.g., purchases) and non-monetary behaviors (e.g., likes, comments, and shares) (Tian and Frank, 2024). Even in the absence of direct purchases, consumer interactions such as feedback and content engagement can significantly influence future sales (Lim et al., 2022). More recently, in the context of live-streaming commerce, Clement Addo et al. (2021) conceptualize live-streaming engagement as real-time interactions focused on relevant topics, including discussions about streamers and products, facilitated by audience actions. Similarly, Luo et al. (2024a) define customer engagement as observable user actions which indicate participation in live streaming. Kang et al. (2021) identify interactivity as a defining feature of live streaming, where customer engagement unfolds in real time through dynamic interactions (Song et al., 2024). Based on the valuable insights from these definitions, in line with the context of this study we have integrated the perspectives of Vivek (2009), Islam and Rahman (2016), Luo et al. (2024a) to reframe the concept of customer engagement for this study; accordingly, we defined customer engagement as the observable, real-time interactive behaviors of consumers in live-streaming commerce that signal their active participation.

Traditionally, customer engagement has been measured using self-reported surveys, primarily because e-commerce platforms were not widespread and obtaining real-time big data was challenging (Vivek et al., 2014). Consequently, earlier research on live-streaming commerce has largely relied on questionnaires to assess customer engagement (e.g., Xue et al., 2020). Recently, many scholars in the live-streaming research field have begun to incorporate real-time data into qualitative research (e.g., Kang et al., 2021). Although questionnaire surveys are widely used, they are limited by respondents' subjectivity and the time-consuming nature of data collection (Kar and Dwivedi, 2020). In contrast, employing real-time data in quantitative research provides a more comprehensive and objective understanding of how streamers' persuasive language combinations influence respondent behavior (Liao et al., 2023).

Previous studies have employed various interaction behavior metrics as key indicators of customer engagement. For instance, Ma et al. (2024) uses like and comments to measure customer engagement in live streaming. Kang et al. (2021) contend that in live-streaming e-commerce, the core dimensions of customer engagement namely consumption, contribution, and creation can be measured using specific indicators such as likes, gift-giving, and comments. Similarly, Luo et al. (2024a) further assess user engagement by emphasizing relationship-building between users and the live-streaming room or broadcasters, incorporating "following" as a key indicator. Although these studies use different measures, they consistently underscore that customer interaction is central to understanding engagement in digital environments. In line with our definition of customer engagement, we have drawn on prior research and contextual considerations to select key indicators that capture interactivity. Consequently, we adopt a comprehensive approach that measures engagement using three variables: likes, comments, and follows.

Table 2 highlights that existing literature on customer engagement in live streaming has primarily focused on drivers such as interactivity (Kang et al., 2021), user trust (Guo et al., 2021), relational bonds (Hu and Chaudhry, 2020), utilitarian and hedonic values (Wongkitrungrueng and Assarut, 2020), live interactions (Xue et al., 2020), speech acts (Ma et al., 2024), and streamer topics (Luo et al., 2024a). Given the inherently social nature of live e-commerce, customer engagement plays a critical role. For instance, it mediates the relationship between streamer performance/behavior and purchase behavior (Ma et al., 2024; Luo et al., 2024b). Moreover, Deng et al. (2021) have

Table 2
Relevant literatures on customer engagement in live streaming e commerce context.

Reference	Context	Customer engagement variable types	Antecedent of customer engagement	Measured method of customer engagement	Research Focus
Hu and Chaudhry (2020)	Live streaming	Dependent Variable	Relational bond (financial, social,and structural bonds), commitment to streamer	Questionnaire	Highlight the social bond between streamer and consumer on customer engagement
Xue et al. (2020)	Live streaming	Dependent Variable	Live interactions (i.e., Personalization, Responsiveness, Entertainment, Mutuality, Perceived control)	Questionnaire	Consider the interaction between streamers and consumers as a key factor influencing customer engagement.
Wongkitrungrueng and Assarut (2020)	Live streaming	Dependent Variable	Customers' perceived value (i.e., Symbolic value), Customer trust	Questionnaire	Highlight the impact of customer trust in live streaming e commerce
Guo et al. (2021)	Live streaming	Dependent Variable	Customer trust (i.e., trust in Broadcasters, Community embers, and products), Swift guanxi	Questionnaire	Highlight the impact swift guanxi in the relationship between customer trust and engagement.
Kang et al. (2021)	Live streaming	Dependent Variable	Interactivity (responsiveness, personalization), Tie strength	Econometric model	Highlight the link between interactivity and customer engagement behavior.
Deng et al. (2021)	Social Media	Dependent Variable	Brand post linguistic styles (emotionality, complexity and informality)	Econometric model	Highlight the role that brand language plays in shaping consumer engagement on social media.
Yu and Zheng (2022)	Live streaming	Mediating Variable	Consumers' perceived luxury values (financial value, functional value, individual value and social value)	Questionnaire	Highlight the impact of perceived luxury values on customer engagement in the context of live streaming luxury shopping
Luo et al. (2024a)	Live streaming	Dependent Variable	Informative topic, Affective topic	Econometric model	Highlight the impact of two types of topics discussed by streamers and examines their influence on user engagement.
Luo et al. (2024b)	Live streaming	Mediating Variable	ELM-informed central (i.e., product information quality, streamer interaction quality, and streamer credibility) and peripheral (i.e., review consistency) routes	Questionnaire	Highlight the importance of informational live streaming commerce elements and their effect on customer engagement and impulsive buying tendency
Tian and Frank (2024)	Live streaming	Dependent Variable	Flow experience	Questionnaire	Focus on flow experience mediates the effects of live streaming features on customer engagement
Ma et al. (2024)	Live streaming	Mediating Variable	Speech act: (Representative act, Expressive act, Commissive act, Directive act)	Econometric model	Focus on the effect of different speech acts on customer engagement and purchase behaviour
This study	**Live streaming**	**Mediating Variable**	**Linguistic persuasive combinations (Reward and exaggeration, Reward and logic, Reward and emotion, Credibility and exaggeration, Credibility and logic, Credibility and emotion)**	**Econometric model**	Highlight the importance of linguistic persuasive combinations of streamer on customer engagement and sales

shown that the linguistic style of social media posts significantly affects engagement, while Ma et al. (2024) demonstrate that streamers can guide consumer participation through their speech act. In fact, variations in linguistic style can capture consumer attention and encourage active participation in discussions (Mubarok et al., 2024). Although previous research has established the mediating role of customer engagement in linking streamer performance to consumer behavior, studies examining the impact of persuasive linguistic strategies on engagement remain scarce. By taking customer engagement as a mediating variable, we can better understand how the persuasive language combination used by live streamers influences overall sales performance, thereby solidifying both the theoretical and practical significance of our study. Furthermore, combining Hovland's persuasion model with Aristotle's rhetorical theory can refine streamers' persuasive language strategies and supports a detailed investigation of how various persuasive linguistic combinations influence consumer behavior. This study reveals the mechanism linking persuasive language combinations with the extent of customer engagement (shown in Table 2), offering novel insights into the literature.

2.3. Electronic words of mouth

Aristotle's distinction between artistic and non-artistic persuasion provides a useful framework for understanding the role of eWOM. Artistic persuasion refers to strategies crafted by the persuader, such as the use of rhetorical appeals, while non-artistic persuasion involves external factors that the persuader cannot create but can utilize to support the argument (Rapp, 2023). E-commerce retailers have recognized the effectiveness of utilizing electronic Word of Mouth (eWOM) as a marketing strategy that greatly influences consumer purchasing decisions (Levy and Gvili, 2024). Hennig-Thurau et al. (2004) describes electronic word of mouth as a message shared online by potential, current, or past customers regarding a product or company, accessible to numerous individuals and organizations through the Internet. In live streaming e commerce, eWOM includes consumers liking, commenting, or sharing behaviour (Chu and Kim, 2011; Chen et al., 2019; Gvili and Levy, 2018). Babić Rosario et al. (2020) found that consumers are driven to engage with eWOM for various reasons, including enhancing their knowledge, minimizing the effort involved in pre-purchase evaluation,

lowering perceived risks (such as with new or costly products or hidden attributes), and alleviating cognitive dissonance. Additionally, some consumers view eWOM as a form of entertainment, like humorous reviews, or encounter it unintentionally, such as through influencers or fear of missing out (FOMO). Marketers encourage consumers to generate eWOM by employing strategies such as nudges, Q&A interactions, community engagement (Eelen et al., 2017), and providing monetary or non-monetary incentives (Ahrens et al., 2013; Levy and Gvili, 2024), Simultaneously, they implement measures to discourage the creation of negative eWOM (Babić Rosario et al., 2020). Studies indicate that eWOM has grown into a crucial factor impacting consumer behavior across multiple online and offline settings (Donthu et al., 2021). Verma and Dewani (2021) identified that the perceived credibility of eWOM is influenced by factors such as content, the communicator, context, and the consumer. This credibility, in turn, impacts consumers' attitudes and perceptions toward the brand (Haq et al., 2024). eWOM can affect product sales by influencing consumer perceptions and evaluations (Chevalier and Mayzlin, 2006; Ni and Cheng, 2024), playing a significant role in decision-making processes, from positive endorsements to cautionary advice (López-López and Parra, 2016). However, there is still limited research on how eWOM moderates consumer engagement and sales in live-streaming contexts.

3. Hypothesis development

3.1. Persuasive language combinations and product sales

The persuasion theories of Aristotle and Hovland, pointing out that linguistic persuasion can influence people's decision-making through cognitive and affective reactions (Luo et al., 2021; Bronstein, 2013; Yang et al., 2018). Persuasive information can change people's thoughts, feelings and actions, especially in purchasing goods and services (McGuire, 2000). Persuasive language therefore tends to be creative and attention-drawing, exploiting language resources to the maximum with catchphrases, emotive words, informal expressions and striking metaphors and comparisons to appeal to the audience (Labrador et al., 2014). Reward-based language typically involves offering incentives, highlighting benefits, or framing a purchase as an advantageous decision for the consumer, which significantly influence product sales (Luo et al., 2021). Streamers often promise both material and non-material incentives, such as coupons, giveaways, or temporary price reductions, to encourage purchases during broadcasts. Reward-focused language taps into extrinsic motivation, where consumers are drawn to the perceived advantages or value offered by a product.

Credibility is an essential factor in persuasive communication, as it lends trustworthiness and reliability to the speaker's message (Hovland and Weiss, 1951). Aristotle's rhetorical appeals pointed out that the persuader obtains the audience's trust in personality traits by showing knowledge and morality. El Baff et al. (2020) argued that incorporating accurate information increases the credibility of content. The credibility of a persuasive message depends more on the "source" than on the content itself (Choi et al., 2005). Credibility-based language includes elements that signal expertise, honesty, and authority (Lin et al., 2021b), which can reassure potential buyers and reduce the perceived risk of making a purchase. During live streams, a streamer may use language that emphasizes the credibility of the product, such as guaranteeing its quality and presenting laboratory test results from reputable authorities. When consumers perceive a message as credible, they are more likely to respond positively and engage in purchasing behavior (Gao et al., 2021).

Exaggerated language will arouse some consumers' interest in the product (Luo et al., 2021). It draws on the Aristotle's rhetorical appealing to the excitement of potential buyers and creating an impression of heightened value. Exaggeration, when used carefully alongside credibility-based language, can further enhance a product's appeal by drawing attention to key features and creating a sense of excitement. In a live stream, the streamer might use phrases such as "very," "first," "oh my God," and "best" to present products to attract consumers' trust, and using exaggerated phrases, such as "the largest in history" and "super cost-effective," to express rewards in the live stream, thus enhancing customers' purchase behavior. This blend allows live streamers to present themselves as both enthusiastic advocates and reliable sources, making the persuasive attempt more compelling. Thus, we propose further hypotheses as follows:

Hypothesis 1a. Persuasive linguistic combinations based on reward and exaggeration are positively related to product sales.

Hypothesis 1b. Persuasive linguistic combinations based on credibility and exaggeration are positively related to product sales.

Linguistic logic is defined by how structured and systematic the language is (Bartlett, 2019). Logic implies that the persuader emphasizes logic in the process of persuasion and proves their point of view using logical arguments (Chen et al., 2024). Logic is achieved when the speech establishes truth (Braet, 1992). The persuader structures the message with key terms that highlight aspects like facts, evidence, expert opinions, and common sense (Aronson et al., 1963). In live streaming, logic refers to how well the streamers' language is structured, coherent, and grounded in facts and theoretical principles (Chang et al., 2018). During live streaming, streamers use different linguistic persuasion modes, such as utilizing logical or rational elements in language expressions to convince consumers. However, when reward-based language is combined with logical appeals, the persuasive effectiveness may become less impactful in a live streaming context. This is because live streaming is a dynamic and fast-paced environment where viewers often rely on quick emotional and instinctive responses rather than prolonged rational deliberation (Luo et al., 2021; Lo et al., 2022). Combining reward (an emotionally driven appeal) with logic (a cognitively driven appeal) may create conflicting cues for the audience. While the reward aspect aims to excite and motivate, according to dual process theory the logical component requires deeper thought and analysis (Chen et al., 2022), potentially slowing down decision-making and reducing the overall persuasiveness of the message.

The use of logical language by live streamers allows users to thoroughly understand product information (Chen et al., 2024). Credible and logical language meets consumers' preferences for obtaining information and helps reduce perceived uncertainty (Shin et al., 2017). In live-streaming e-commerce, where consumers often seek detailed and trustworthy information before committing to a purchase (Liu et al., 2023b), the integration of credibility and logic can create a powerful persuasive effect that enhances consumer confidence and encourages sales. Thus, we propose further hypotheses as follows:

Hypothesis 1c. Persuasive linguistic combinations based on reward and logic are negative related to product sales.

Hypothesis 1d. Persuasive linguistic combinations based on credibility and logic are positively related to product sales.

Emotional language involves using speech to evoke feelings in the audience (Braca and Dondio, 2023). This strategy entails the persuader stimulating the audience's emotions to foster empathy and influence their rational decision-making (Lin et al., 2021). Viewers can observe and respond to a streamer's emotions through their linguistic style (Du et al., 2011) which in turn may drive purchase intentions (Lu and Chen, 2021). Emotional appeals can create a connection with consumers, making them feel understood and valued, and can evoke responses (Ma et al., 2024; Gao et al., 2024). We propose that combining reward-based language with emotional appeals enhances the persuasiveness of the message. The reward aspect stimulates the consumer's motivation, while the emotional component strengthens the bond and creates a sense of urgency or desire. This dual approach leverages both the rational desire for benefits and the emotional drive to experience positive feelings, enhancing the overall persuasiveness and effectiveness of the communication.

Emotional processes have been shown to contribute to persuasive outcomes (Petty and Briñol, 2014; Breves, 2023). Credible message pair with urgent phrases, such as "quick grab before is gone" and "our satisfied customers feel secure and excited about their choice," trigger strong emotional reactions and encourage immediate consumer action, thereby stimulating purchasing behavior (Lin et al., 2021). Thus, we propose further hypotheses as follows:

Hypothesis 1e. Persuasive linguistic combinations based on reward and emotion are positively related to product sales.

Hypothesis 1f. Persuasive linguistic combinations based on credibility and emotion are positively related to product sales.

3.2. Persuasive language combinations and customer engagement

Among the four basic elements of Hovland's persuasion model mentioned above, the persuader, persuasive information, and persuasion situation are external stimuli for attitude changes. Swartz (1976) believed that exaggeration is a "construction" of reality that can enable people to focus their attention, thus downplaying the description and bringing the audience into the speaker's perspective in a powerful way. Kim and Kim (1999) noted that a company can use an exaggerated language style to describe the quality of products and forecast new products to increase the probability of customers returning. Chakraborty and Swinney (2021) proposed that investors know less about product quality than financiers; this leads to information asymmetry, which gives financiers the opportunity to exaggerate product quality. By exaggerating the quality of products, financiers persuade investors to participate in crowdfunding activities. Therefore, in an interaction, exaggerated language will make consumers pay attention to what information sender says. Thus, we suggest that in the live streaming situation, when a streamer uses exaggerated rhetoric to describe rewards and products, it can create strong customer engagement.

The streamer explains the products to consumers through pictures and language, allowing consumers to understand the value and practicality of the products online (Ma and Mei, 2019). The streamer can also introduce live streaming events through language to increase entertainment and provide consumers with more entertaining experience in the process of engagement. Therefore, the speech of the streamer will affect the consumer's cognition, which in turn affects their engagement behavior. Messages from credible sources build trust and promote deeper message processing, which in turn enhances customer engagement (Wang and Scheinbaum, 2018; Lee and Koo, 2016). According to Craig and Blankenship (2011), the use of intense or extreme language can increase both persuasion and perceived credibility. Additionally, streamers viewed as knowledgeable stimulate greater interaction (Li et al., 2023), making the combination of credibility and strategic exaggeration an effective approach for engaging consumers. Xue et al. (2020) suggested that the streamer's speech is a parasocial behaviour for customers, which can positively decrease the perceived risk and psychological distance, thus promoting the society to improve customer engagement. We thus propose further hypotheses as follows:

Hypothesis 2a. Persuasive linguistic combinations based on reward and exaggeration are positively related to customer engagement.

Hypothesis 2b. Persuasive linguistic combinations based on credibility and exaggeration are positively related to customer engagement.

Logical style in discourse primarily reflects a speaker's awareness of logical structure, helping to avoid misinformation. The concept of 'appeal to logic' involves conclusions drawn through rational reasoning and is an effective strategy for dispelling rumors (Gentes and Selker, 2013). Streamers often incorporate logical elements in product information or promotional details, aligning with Aristotle's three modes of persuasion. For instance, when offering rewards during live streaming, streamers might use phrases like "significant reward" or "discount." While logical clarity can aid comprehension, it may also lead to a sense of detachment (Hong et al., 2020). Explaining the rules for rewards in purely logical terms can sometimes prompt viewers to downvote the streamer or exit the session. Therefore, we posit that logical language use during reward announcements could negatively affect customer engagement.

Conversely, when introducing products, streamers often highlight various attributes, such as type and performance, to bolster audience confidence. This approach can encourage viewer engagement, including likes and comments, suggesting that logical language when explaining product details positively impacts trust and customer interaction (Moe, 2003). Thus, we propose further hypotheses as follows:

Hypothesis 2c. Persuasive linguistic combinations based on rewards and logic are negatively related to customer engagement.

Hypothesis 2d. Persuasive linguistic combinations based on credibility and logic are positively related to customer engagement.

Previous marketing communication studies have found that emotional information (i.e., the use of emotional words when conveying information) stimulates greater cognitive engagement (for example, attitude) (Kissler et al., 2007) and psychological change (Berger and Milkman, 2012) and improve the engagement of information receivers (Joyce and Kraut, 2006) as well as their sharing behaviors (Kim and Johnson, 2016; Peters et al., 2009). In addition, previous studies have claimed that emotions embedded in brand-related content on Facebook will affect consumers' engagement behavior such as likes, comments (Kim and Johnson, 2016). Brand posts with emotional content improve engagement behavior (Coursaris et al., 2016; Lee et al., 2018). Emotional information being conveyed by a streamer in a live stream can affect the attitude, mood, and behavior of the consumer, thereby achieving the purpose of persuasion. The streamer might use phrases such as "I will be happy if everyone is happy" and obtain the audience's approval by creating an emotional group atmosphere. Especially when awarding rewards, the streamer might use the phrase "this is the reward I have won for you personally" and repeatedly use friendly phrases, such as "sisters" and "family" to mobilize consumer empathy and enhance consumer engagement when explaining the goods. The blend of credibility and emotion can deepen customer engagement by fostering trust while simultaneously making the interaction relatable and heartfelt. When the streamer frequently uses emotional words to persuade, it stimulates consumer engagement behavior. Engaged viewers are more likely to react, ask questions, and remain involved in the broadcast when they feel both emotionally connected and assured of the speaker's trustworthiness (Li et al., 2024). We thus propose another hypothesis as follows:

Hypothesis 2e. Persuasive linguistic combinations based on rewards and emotion are positively related to customer engagement.

Hypothesis 2f. Persuasive linguistic combinations based on credibility and emotion are positively related to customer engagement.

3.3. Customer engagement and product sales

Parhankangas and Renko (2017) showed in a study on live streaming marketing that customer engagement is a precursor to consumers' greater willingness to buy. Hsieh and Chang (2016) also found a positive association between customer engagement and purchase intentions. Luo et al. (2024b) showed that customer engagement is positively related to impulse buying. Kaveh et al. (2021) believed that attracting customers is positively associated with purchase intentions, which is mediated by customer engagement behavior (such as commenting, liking, and sharing). The sales of products reflect the attractiveness and success of the streamer and live streaming platform. Therefore, live streaming needs to concentrate on customer engagement, and the degree of customer engagement has an important association with sales. We thus

propose a new hypothesis as follows:

Hypothesis 3. There is a positive correlation between customer engagement and product sales.

Research indicates that customers prefer video content over text and images, as visual cues help direct their attention to both the functional and emotional aspects of a product (Kahn, 2017). At the same time, Zheng et al. (2022) and Li et al. (2022) have shown that comments serve as strong positive predictors of purchase intention in live-streaming commerce. When consumers have engaging and co-created experiences online, they are more likely to participate actively, maintain their involvement, and seek notifications and updates (Tafesse and Wien, 2018). According to Kaveh et al. (2021), attracting customers is positively correlated with purchase intentions, with customer engagement behaviors, such as attention and praise, mediating this relationship. The streamer's persuasive language strategy significantly impacts product sales in live streaming (Luo et al., 2021). Given the communicative role of the streamer's language, it is likely to influence customer engagement and, consequently, their purchasing behavior (Ma et al., 2024). Therefore, we posit that customer engagement mediates the relationship between product sales and persuasive linguistic combinations.

The combination of reward-based language and exaggeration evokes excitement and urgency. This combination captures viewers' attention and fosters high levels of engagement (Swartz, 1976). By using phrases such as "limited-time best deal" or "biggest discount ever," streamers generate excitement that motivate viewers to interact, comment, and ultimately purchase. In this context, customer engagement acts as an intermediary, transforming initial interest into concrete sales.

Aristotle's ethos highlights that a speaker's perceived trustworthiness can be enhanced by confidently stated claims (Baumrin, 1986). Statements like "This is the top-rated product in its category" create excitement while maintaining message integrity. Combining credibility with a touch of exaggeration can bolster trust and maintain viewer enthusiasm. Customer engagement mediates this effect by reinforcing viewers' trust and excitement, prompting interaction and converting positive emotions into sales. Thus, we propose further hypotheses as follows:

Hypothesis 4a. Customer engagement has a mediating effect between product sales and persuasive linguistic combinations based on rewards and exaggeration.

Hypothesis 4b. Customer engagement has a mediating effect between product sales and persuasive linguistic combinations based on credibility and exaggeration.

While reward-based language stimulates interest and appeals to consumer motivations, logic-based language provides structured, rational arguments and evidence to support the claims being made (Aristotle, 1954; Chang et al., 2018). The combination of these linguistic styles can present a dual persuasive effect where emotional appeal is paired with cognitive reasoning. However, in the live streaming context, where quick and impulsive reactions are common, customer engagement becomes crucial in bridging the gap between persuasion and purchasing behavior (Luo et al., 2024a).

Combining credibility with logic creates a rational and trustworthy message that appeals to the audience's sense of logic and reason. This approach is particularly suited for viewers who are looking for informative and evidence-based arguments before making a purchase decision (Liu et al., 2023b). Customer engagement becomes essential in mediating this combination to ensure that it results in effective persuasion and higher sales. Thus, we propose further hypotheses as follows:

Hypothesis 4c. Customer engagement has a mediating effect between product sales and persuasive linguistic combinations based on rewards and logic.

Hypothesis 4d. Customer engagement has a mediating effect between product sales and persuasive linguistic combinations based on credibility and logic.

The fusion of reward-based language with emotional appeals can amplify the overall persuasive effect by appealing to both the viewer's desires and emotions. Fredrickson and Branigan (2005) suggest that positive emotions lead people to take actions that help refine skills, solve problems, or achieve goals. Emotional messages generate higher behavioral engagement compared to other message types (Tafesse and Wien, 2018; Dolan et al., 2019). When viewers are emotionally invested and perceive potential benefits, their engagement often translates into stronger purchasing behavior.

Credibility-based language indicates knowledge and expertise, reinforcing perceived credibility (Syrdal et al., 2023). Emotional content promotes positive feelings such as amusement and contentment, fostering creativity, compassion, cooperation, and increased attention while promoting social connections (Fredrickson and Branigan, 2005). Emotional appeals may also drive social interaction (Lee and Koo, 2016). When used together, these elements can engage the audience effectively by ensuring that the message resonates on both a rational and emotional level. Customer engagement can play a significant role in mediating this relationship, facilitating the transition from viewer interest to actual purchasing behavior. Thus, we propose further hypotheses as follows:

Hypothesis 4e. Customer engagement has a mediating effect between product sales and persuasive linguistic combinations based on rewards and emotion.

Hypothesis 4f. Customer engagement has a mediating effect between product sales and persuasive linguistic combinations based on credibility and emotion.

3.4. Electronic word of mouth (eWOM)

EWOM is the evaluation of a product or service posted by consumers online (Abbasi et al., 2023). EWOM plays an important role in live streaming. During live streams, customers usually get a sense of product quality and recognize corporate identifiers through eWOM for products and services they have not experienced (Wang et al., 2022). Thomas and Saenger (2017) argued that eWOM works faster when institutions and individuals have higher perceived quality and better-known brand images in the past. Seller reputation is an important factor affecting consumers' purchase choices (Shareef et al., 2008). In addition, retail websites updates sellers' reputation scores in real time through online reputation feedback systems. The buyer's feedback covers the consistency between the product and the description, the seller's service attitude and the seller's delivery speed. Wang et al. (2019) argued that consumers' engagement in brand eWOM communication on social media platforms depends on consumers' engagement with and attitude towards a brand through social interaction. Therefore, in the context of live streaming with goods, eWOM is represented by the combined rating of consumers in the live streaming room, which is an objective non-artistic persuasion factor. For the audience, eWOM in the live streaming room is a low-cost and effective way to obtain consumer trust, and it plays an important role as a source of information. Therefore, after the audience has developed certain engagement behaviors, non-artistic persuasion factors can improve the customer's sense of trust and thus motivate purchase behaviors. That is, the eWOM of the live streaming room will enhance the positive relationship between customer engagement and product sales. We thus propose another hypothesis as follows:

Hypothesis 5. EWOM plays a positive moderating role between customer engagement and product sales.

Our research model is shown in Fig. 1.

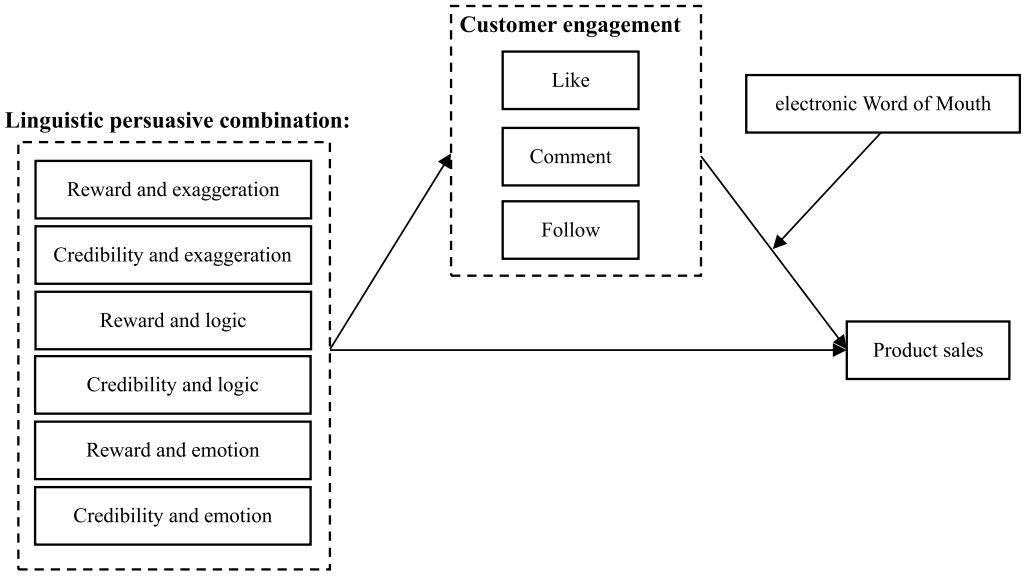


Fig. 1. Conceptual model.

4. Data and methodology

4.1. Data collection

This study selected the live streaming platform TikTok as the data source. TikTok Live was founded in 2016 and has become one of the most popular streaming media platforms for live video games in China. It mainly focuses on short video games and e-commerce live streaming but also provides education, entertainment, and other streaming media content. As of August 2020, TikTok had more than 600 million active users every day, and it continues to grow at a high speed. Since this paper mainly studies the impact of streamer speech on customer behavior, to avoid certain research limitations in selecting streamers with high or low popularity, we chose the beauty live streaming of different levels of fans on TikTok as the research object, namely Zhu Zixiao (number of fans: 14 million), Shuchang (number of fans: 8.8 million), Sister Pandora (number of fans: 460 thousand) Doudou in Hainan (number of fans: 170 thousand), and 10 other streamers. The live network celebrities selected in this study were recorded in TikTok live streaming rooms from November 1, 2022, to November 15, 2022, in real time. This time was the occasion of the Double 11 Shopping Festival, which involves large-scale centralized promotion over live streams. While introducing the products, the streamer will also offer rich promotional rewards to the audience, which will help us study the speech acts of the streamer, who uses different linguistic persuasive combination when handing out rewards and explaining the products.

The data collection was divided into two parts. In the first part, we captured live streaming video and used Automatic Speech Recognition (ASR) technology to convert voice to text. We collected 56,032 streamer words, of which 49,404 remained after cleaning and sorting out the text. We then built three text classification models using the in-depth learning technology of the Baidu EasyDL platform. The first text classification model manually divided 5000 streamer words into two categories: credibility and rewarding. Through training, an Application Programming Interface(API) classification interface with an accuracy rate of up to 95 % was obtained. The API interface of the trained binary text classification model was used to classify the remaining 44,404 unlabeled words using AI smart tags, and 11,660 reward tags and 32,744 trusted tags were obtained. The second model was used to extract 2000 words from 32,744 trusted labels and label them manually based on the use of exaggeration, logic, and emotion. After training, an API classification interface with an accuracy rate of 92 % was obtained. The third model was used to extract 2000 words from 11,660 reward tags to manually annotate exaggeration, logic, and emotion and obtain an API classification interface with an accuracy rate of 93 %. Finally, the API interfaces of the last two text classification models were used to classify the remaining 30,744 trust-related texts and 9660 award-related texts with AI smart tags, and a total of 40,404 AI-tagged words were obtained.

The process of manual annotation was inspired by the annotation methods proposed by Hidey (Hidey et al., 2017). Firstly, Table 3 according to the relevant theoretical bases of the variables, the labeling standards of different variables were determined. Secondly, two analysts randomly selected 600 words to carry out three rounds of independent labeling according to the labeling standards, subsequently exchanging the labeling results and analyzing the differences. After three rounds of labeling and discussion, the final labeling standards were obtained. The final standards are shown in Table 3. We found five students to train the labeling standard scheme, and then they labeled the remaining 8100 texts. Two analysts randomly selected 500 of the 8100 texts marked by the five students for independent labeling and checked the labeling effect. The coincidence rate reached more than 95 %. Therefore, we believe that the labeled text can be used to train the AI model in text classification.

In the second part, we used the Zhigua data platform. Through observation of nearly 10,000 live broadcast booths with goods on TikTok from July to September 2022, we found that the average stay time of the audience in the live broadcast booths with goods was about 2 min and 30 s. Therefore, based on the unit of 2 min and 30 s, combined with

Table 3
Manual classification standard.

Classification	Defining standard
Credibility	Product introduction, product attributes, detailed product information, product use effect and experience, product certificate report, quality assurance, etc.
Reward	Gift in live stream room (lucky bag, free delivery, etc.), product discount (cheap, discount), spiritual report (perceived cheap)
Exaggeration	Exaggerated or extreme description (super, too, special, perfect, most, etc.), interjections (Oh my god, wow, etc.)
Logic	Logical deduction, causal relationship, rational argument and evidence
Emotion	Emotional shock(upload the purchase link and urge audience to snap up), ask for gratitude (ask to follow the studio, join a fan group, like, etc.), get empathy (describe putting a lot of effort into the product, etc.)

the sales data collected on Zhigua and the combined data of streamer's persuasive linguistic combinations, we found that the sales data 5 min before the opening of the live stream were generally none, due to pre-broadcast preparations and initial audience buildup. Therefore, to ensure the reliability of the results, we chose to remove this part of the data in the final study—that is, the corresponding streamer's speech data was also removed. The text data of 36,166 persuasive linguistic combinations used by streamers were obtained for the final study. Finally, the streamer text data and other data were summarized to obtain a total of 649 valid data for model research.

4.2. Build model

Based on Hovland's persuasion model and Aristotle's rhetorical triplet theory, this paper establishes a persuasive linguistic combination model including six persuasive linguistic combinations: rewards and exaggeration, rewards and logic, rewards and emotion, credibility and exaggeration, credibility and logic, and credibility and emotion. Based on the persuasion model, we studied the influence of these combinations on customer engagement and product sales in a live streaming e-commerce environment. To test the influence of the combinations on customer engagement and sales volume in the live stream, the following basic econometric model was constructed:

$$\text{Model 1}: \mathrm{Ln(Sales)_i} = a_0 + a_1 \mathrm{RE_i} + a_2 \mathrm{RL_i} + a_3 \mathrm{LnREM_i} + a_4 \mathrm{CE_i} + a_5 \mathrm{CL_i} + a_6 \mathrm{CEM_i} + a_7 \mathrm{controls_i} + \varepsilon_i \quad (1)$$

$$\text{Model 2}: \mathrm{Ln(CEN)_i} = a_0 + a_1 \mathrm{RE_i} + a_2 \mathrm{RL_i} + a_3 \mathrm{LnREM_i} + a_4 \mathrm{CE_i} + a_5 \mathrm{CL_i} + a_6 \mathrm{CEM_i} + a_7 \mathrm{controls_i} + \varepsilon_i \quad (2)$$

$$\text{Model 3}: \mathrm{Ln(Sales)_i} = a_0 + a_1 \mathrm{RE_i} + a_2 \mathrm{RL_i} + a_3 \mathrm{LnREM_i} + a_4 \mathrm{CE_i} + a_5 \mathrm{CL_i} + a_6 \mathrm{CEM_i} + a_7 \mathrm{Ln(CEN)_i} + a_8 \mathrm{controls_i} + \varepsilon_i \quad (3)$$

$$\text{Model 4}: \mathrm{Ln(Sales)_i} = a_0 + a_1 \mathrm{Ln(CEN)_i} + a_2 \mathrm{Ln(CEN)_i}{}^{*}(\mathrm{eWOM})_i + a_3 (\mathrm{eWOM})_i + a_4 \mathrm{controls_i} + \varepsilon_i \quad (4)$$

$Ln(Sales)_i$ is the dependent variable of the model, representing the sales volume of the live stream at time "i." CEN (customer engagement) was measured from the three dimensions of likes, comments, and follows. Due to the large variance between these three dimensions, we took the logarithms of their sums. RE_i, RL_i, and REM_i represent the number of exaggerated, logical, and emotional statements used by the streamer when awarding awards during this time period. CE_i, CL_i, and CEM_i represent the number of exaggerated, logical, and emotional statements used by the streamer when explaining goods in this period. $eWOM_i$ is the public praise score of the live stream in this period, while $Ln(CEN)_i{}^{*}(\mathrm{eWOM})_i$ represents the cross-comparison of eWOM and customer engagement. Controls represent the control variable, composed of the frequency of the live stream and the number of fans. ε_i is the random error related to observation "i." The system consists of four interrelated equations: Model 1 represents the influence of explanatory variables on sales, Model 2 explains the influence of explanatory variables on customer engagement, Model 3 represents the influence of explanatory variables and customer engagement on sales, and Model 4 represents the influence of moderator variable We estimated the equation in two steps to test each hypothesis. First, Models 1, 2, and 3 were tested at the same time to test the intermediary effect of the model, and then Model 4 was used to verify the moderating effect of eWOM.

5. Data analysis and results

5.1. Descriptive statistics

We used Stata 17.0 for the data analysis. Table 4 provides the descriptive statistics of 649 pieces of valid data. The results show that the average sales volume of the streamers in the live streaming delivery was 1,382, and the streamer with the largest sales volume sold 105,600 items in 2 min and 30 s. At the same time, it was found that the streamers preferred to use the persuasive linguistic combination of credibility and logic when selling goods. Table 5 shows the correlation between variables. In addition, we analyzed the collinearity of the variables. The variance expansion factor related to all variables was less than 10 (the average value is 2.38), so the problem of multicollinearity was ruled out.

5.2. Hypothesis testing

The results from Model 1 reveal varied impacts of different persuasive linguistic combinations on product sales (Table 6). Reward and exaggeration positively impact product sales ($a1 = 0.209$, $p < 0.001$), supporting H1a. Credibility and exaggeration ($a2 = 0.053$, $p < 0.05$) positively impact sales, supporting H1b. However, reward and logic ($a3 = -0.016$, $p > 0.05$), and reward and emotion ($a5 = 0.046$, $p < 0.05$) are non-significant, so H1c and H1e are not supported. The analysis for H1d, which proposed that the combination of credibility and logical language would positively affect sales, was not supported. Instead, the results showed that credibility and logic negatively affected sales ($a4 = -0.061$, $p < 0.001$). However, credibility and emotion did not significantly support H1f ($a6 = 0.002$, $p < 0.05$). These results suggest that emotional and exaggerated language styles are generally more effective in driving sales in live streaming e-commerce.

The results for customer engagement impact are shown in Table 6 and Fig. 2. Model 2 shows the multiple regression results of the combination of the streamer's speech persuasion behaviors and customer engagement. H2a ($a1 = 0.099$, $p < 0.001$), H2b ($a2 = 0.037$, $p < 0.05$), H2e ($a5 = 0.087$, $p < 0.001$), and H2f ($a6 = 0.025$, $p < 0.05$) indicate

Table 4
Descriptive statistics.

Variable	N	Mean	p50	SD	Min	Max
Sales	649	1382	32	5099	0	105600
RE(reward and exaggeration)	649	2.889	2	3.126	0	14
RL(reward and logic)	649	5.750	4	4.656	0	23
REM(reward and emotion)	649	4.339	3	3.647	0	20
CE(credibility and exaggeration)	649	4.758	4	3.507	0	30
CL(credibility and logic)	649	27.49	27	11.17	1	58
CRE(credibility and emotion)	649	10.50	10	5.630	1	31
CEN(customer engagement)	649	56981	972	112195	2	884442
eWOM	649	4.550	4.470	0.124	4.410	4.720
Fans	649	560.8	47.10	598.2	17.30	1401
Frequency	649	36.58	36	10.45	14	49

Table 5
Correlation among the main variables.

	Sales	RE	RL	REM	CE	CL	CEM	CEN	eWOM	Fans	Frequency
Sales	1										
RE	0.364***	1									
RL	0.239***	0.541***	1								
REM	0.217***	0.421***	0.528***	1							
CE	0.147***	0.178***	0.180***	0.074*	1						
CL	−0.292***	−0.585***	−0.563***	−0.611***	−0.277***	1					
CRE	−0.172***	−0.264***	−0.152***	0.129***	−0.0170	−0.0140	1				
CEN	0.309***	0.547***	0.442***	0.474***	0.099**	−0.536***	−0.245***	1			
eWOM	0.328***	0.610***	0.497***	0.433***	0.323***	−0.560***	−0.0310	0.418***	1		
Fans	0.365***	0.721***	0.641***	0.454***	0.356***	−0.646***	−0.291***	0.608***	0.739***	1	
Frequency	−0.120***	−0.291***	−0.390***	−0.456***	−0.193***	0.467***	−0.085**	−0.385***	−0.257***	−0.505***	1

Table 6
Regression analysis.

	Model 1	Model 2	Model 3
	Ln Sales	Ln CEN	Ln Sales
RE	**0.209*** **	**0.099*** **	**0.184*** **
	(0.033)	**(0.027)**	**(0.033)**
CE	**0.053***	**0.037***	**0.043***
	(0.021)	**(0.017)**	**(0.021)**
RL	−0.016	−0.011	−0.013
	(0.022)	(0.017)	(0.021)
CL	**−0.061*** **	**−0.070*** **	**−0.043*** **
	(0.009)	**(0.008)**	**(0.010)**
Ln REM	0.046	**0.087*** **	0.023
	(0.026)	**(0.021)**	(0.026)
CEM	0.002	**0.025***	−0.004
	(0.014)	**(0.011)**	(0.014)
Ln CEN			**0.256*** **
			(0.048)
Ln Fans	0.817***	1.395***	0.459***
	(0.072)	(0.058)	(0.097)
Ln Frequency	−0.293	−2.121***	0.251
	(0.236)	(0.190)	(0.252)
cons	1.335	8.241***	−0.778
	(1.102)	(0.889)	(1.149)
N	649.000	649.000	649.000
R^2	0.670	0.872	0.684
R_a^2	0.666	0.870	0.680

Standard errors in parentheses.
*p < 0.05, **p < 0.01, ***p < 0.001.

that rewards and exaggeration, credibility and exaggeration, rewards and emotion, and credibility and emotion had positive impacts on customer engagement. H2a, H2b, H2e, and H2f were thus supported. Credibility and logic had a negative impact on customer engagement (a4 = −0.070, p < 0.001), meaning that H2d was not supported. The combination of rewards and logic was not related to customer engagement, meaning that H2c was not supported.

We turn now to the mediation effect. According to the distribution test results, the combinations of rewards, exaggeration, and product sales (a1 = 0.209, p < 0.001), rewards, exaggeration, and customer engagement (a1 = 0.099, p < 0.001), and customer engagement and product sales (a7 = 0.256, p < 0.001) were significant. Therefore, it indicated that H3 and H4a were supported, and H4a was part of the intermediary (the intermediary effect accounted for 12 %, and the direct effect was 88 %). Credibility, exaggeration, and product sales volume (a2 = 0.053, p < 0.05), credibility, exaggeration, and customer engagement, (a2 = 0.037, p < 0.05), and customer engagement and product sales volume (a7 = 0.256, p < 0.001) were all significant, so we concluded that H4b was supported as part of the intermediary (the intermediary effect accounted for 19 %, and the direct effect was 81 %). Rewards and logic have nothing to do with product sales and customer engagement, so H4c was not supported. Credibility, logic, and product sales (a4 = −0.061, p < 0.001), credibility, logic, and customer participation (a4 = −0.070, p < 0.001), and customer engagement and product sales (a7 = 0.256, p < 0.001) were significant. Thus, H4d was supported as part of the intermediary (the intermediary effect accounted for 30 %, and the direct effect was 70 %). Rewards and emotion are related to customer engagement, and customer engagement is related to product sales, so customer engagement has a completely intermediary relationship between rewards, emotion, and product sales, meaning that H4e was supported. Credibility and emotion are related to customer engagement, and customer engagement is related to product sales. Therefore, customer engagement had a completely intermediary relationship between credibility, emotion, and product sales, and H4f was supported.

Model 4 in Table 7 and Fig. 2 shows the moderating effect of live streaming eWOM on the relationship between CEN and product sales. The results showed that the cross-comparison between eWOM and customer engagement in the live streams (a2 = 1.529, p < 0.001) was significant. This shows that public praise in a live stream enhances the relationship between customer engagement and product sales, meaning that H5 was supported.

6. Conclusion

6.1. Discussion

This paper investigates the impact of persuasive linguistic combinations on customer engagement and product sales from the perspective of a streamer's discourse in live streaming. It also analyzes the moderating effect of electronic word of mouth (eWOM) on the relationship between customer engagement and product sales. The results are presented in Tables 6 and 7

The results indicate that persuasive language strategies combining rewards and exaggeration, and credibility and exaggeration positively affect customer engagement and product sales, emphasizing the effectiveness of exaggeration in live-streaming e-commerce. Although excessive use of exaggeration can risk sounding inauthentic and lead to negative sales outcomes (Luo et al., 2021), when combined with the reward-based and credibility-based language, it can reinforce the desirability of a product and persuade consumers to make a purchase. The success of combining credibility and exaggeration depends on maintaining a balance where exaggeration is perceived as part of a trustworthy narrative rather than a misleading tactic.

In contrast, the combination of reward and logic did not significantly impact customer engagement or product sales. This finding contradicts Li et al. (2023) and Luo et al. (2021), who reported that logical persuasive language in live streaming could positively or negatively influence sales. One possible explanation is that audiences, as beneficiaries, may not be interested in detailed, logical descriptions of rewards. Researchers have suggested that the emotions conveyed by streamers significantly influence consumer behavior in livestream e-commerce (Lu and Chen, 2021). Instead, they seem to prefer intuitive, immediate experiences delivered through exaggerated and emotional

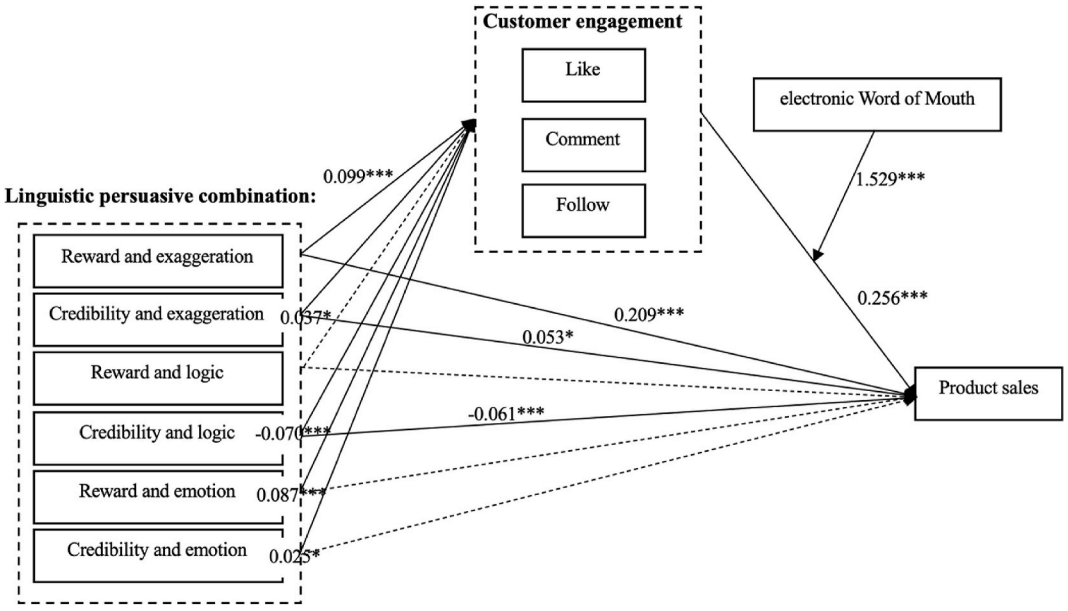


Fig. 2. Model testing results.

Table 7
Regression analysis.

	Model 4
	Ln Sales
Ln CEN	0.140**
	(0.044)
Ln CEN*eWOM	**1.529*** **
	(0.315)
eWOM	2.368**
	(0.800)
Ln Frequency	−0.071***
	(0.013)
Ln Fans	0.002***
	(0.000)
cons	−6.991
	(3.576)
N	649.000
R^2	0.733
R^2_a	0.731

Standard errors in parentheses.
*p < 0.05, **p < 0.01, ***p < 0.001.

language. Additionally, the use of credibility and logic was found to negatively impact customer engagement and product sales, suggesting that overly rational and detailed explanations can deter engagement and reduce sales. This result aligns with Tellis et al. (2019), showing that while logical information can build trust (Chen et al., 2023), it may be perceived as dry and unengaging, ultimately reducing consumer interest and purchase likelihood. This may be due to the nature of the products studied—beauty products, where streamers often rely on professional, objective knowledge that overlaps with the audience's existing understanding, leading to disinterest in repeated explanations. Repeating product details for new viewers can also cause fatigue among the original audience, diminishing engagement and purchase intent.

The findings also show that reward and emotion effectively enhance customer engagement by making viewers feel excited and connected to the content. This aligns with Luo et al. (2024b) but contrasts with Luo et al. (2021) who noted that emotional language positively impacts sales. Luo et al. (2024b) argued that consumers may engage with a streamer's emotional topics. However, emotional language and the promise of rewards alone may not be sufficient to prompt immediate purchases, as viewers sometimes need to interact with the streamer to obtain rewards. This lack of direct impact on sales suggests that while reward and emotion combinations foster interest and interaction, additional persuasive elements may be necessary to drive conversions. Similarly, credibility and emotion proved effective in building an emotional connection with viewers, leading to greater engagement. Previous research has shown that streamers' credibility (Luo et al., 2024a) combined with emotional language can strengthen audience connections, contributing to customer engagement (Syrdal et al., 2023). However, despite these advantages, the combination did not directly increase sales, indicating that while credibility and emotion enhance viewer trust and connection, they may not provide the immediate incentive needed for purchases. The results further underscore the mediating role of customer engagement. Consistent with previous studies (Ma et al., 2024), this research confirms that customer engagement is a crucial antecedent to product sales. Specifically, customer engagement was found to partially mediate the relationship between rewards and exaggeration, credibility and exaggeration, and credibility and logic with product sales. In contrast, it fully mediated the relationships between rewards and emotion, credibility and emotion, and product sales. However, there was no mediating effect for the relationship between rewards and logic and product sales, as H4b was not supported. This suggests that when streamers use highly logical language to introduce products, customer engagement may decrease, potentially due to information overload (Zhang et al., 2024). In addition, excessive detail or contrasting elements can lead to disengagement or skepticism among viewers, diminishing the persuasive impact of the language and reducing the likelihood of purchase. The study suggests that the combination of rewards and logic does not effectively enhance customer engagement or drive product sales.

Furthermore, electronic word of mouth in live streaming plays a positive moderating role between customer engagement and product sales. It can shape consumer perceptions and evaluations, thereby influencing purchasing outcomes (Wang et al., 2018; Upadhyay and Tripathi, 2023). Additionally, a large online audience can further amplify sales potential. Observation learning theory suggests that consumers frequently base their purchasing choices on observed buying behaviors, even if they do not fully comprehend the reasons behind others' decisions (Lu et al., 2021). Consequently, in live-streaming e-commerce context, the feedback and actions of other viewers can significantly affect consumer purchasing behavior.

6.2. Theoretical implications

This study makes several important theoretical contributions to the fields of persuasion theory, rhetorical analysis, and live-streaming e-commerce. First, we address a significant gap in the literature by systematically classifying persuasive linguistic combinations. While prior research has examined individual persuasive elements, such as credibility, logic, and emotion (Li et al., 2023; Luo et al., 2021), few studies have explored how these elements interact in combinations to influence customer engagement and sales. Unlike static forms of persuasion, live-streaming involves dynamic, real-time interactions between streamers and audiences. Streamers often blend different rhetorical techniques to adapt to audience responses, maximize engagement, and drive sales. By integrating Aristotle's Rhetorical Theory and Hovland's Persuasion Model, we highlight how these combinations align with the dual objectives of building trust and creating excitement. This approach provides a more nuanced understanding of how different rhetorical strategies work together to achieve persuasive goals. This contribution underscores the importance of understanding how complex rhetorical strategies influence audience behaviors in high-engagement settings. While Aristotle's Rhetorical Theory has been widely applied in conceptual analyses of live-streaming contexts, previous studies have used methods such as manual coding, computer coding, statistical analysis, and scales to analyze the textual features of language in live streaming e-commerce fields (Luo et al., 2021; Li et al., 2023), the persuasive language of textual features has not been comprehensively explained and discussed at the theoretical level and have rarely been used in an empirical approach to validate these concepts. Our study bridges this gap by leveraging objective, real-world data and employing econometric modelling to examine the effects of different linguistic combinations on customer engagement and product sales. This methodological contribution enhances the applicability of rhetorical theory in live-streaming e-commerce and offers a robust framework in the field.

Therefore, the persuasive linguistic combinations used by streamers, theoretical analysis framework, and measurement methods proposed in this paper comprise an attempt to further investigate the persuasive effects of language.

Second, this paper introduces the variable of customer engagement into the research on verbal persuasion in live marketing and verifies its important intermediary role. While prior research has explored the direct effects of linguistic strategies on purchase behavior, the mechanisms by which customer engagement bridges linguistic strategies and sales outcomes remain underexplored (Luo et al., 2021; Li et al., 2023). Our findings demonstrate how different persuasive linguistic combinations foster engagement behaviors, such as liking, commenting, and following, which in turn drive product sales suggesting that customer engagement plays an important intermediary role between the five verbal persuasion combinations and product sales, except in relation to rewards and logic.

Finally, we extend the application of Aristotle's distinction between artistic and non-artistic persuasion by introducing electronic word-of-mouth (eWOM) as a moderating factor. Most previous studies have only discussed the role of artistic persuasion, ignoring the impact of non-artistic persuasion factors, resulting in one-sided and unsystematic research conclusions (Al-Momani, 2014; Chu et al., 2014; Tirdatov, 2014). This study interpreted persuasive linguistic combinations as artistic persuasion in the delivery process of live streaming and positioned eWOM as a non-artistic persuasion element that influences the relationship between customer engagement and product sales. By analyzing how external credibility sources, such as customer reviews and feedback, amplify the effectiveness of linguistic strategies, we expand the theoretical scope of eWOM in live-streaming contexts. This integration broadens the understanding of how non-artistic persuasion elements interact with artistic language to shape consumer attitudes and behaviors.

6.3. Practical implications

The results of this research provide practical insights for live-streaming platforms, managers, and streamers, emphasizing the use of strategic language as a significant tool for driving customer engagement and boosting product sales. It is essential for managers and streamers to recognize the pivotal role of persuasive language in the context of live streaming. The categorization of language into credibility-focused and reward-based types offers a structured approach for streamers to customize their communication techniques to align with their audience's preferences. For consumers who prioritize trust, language that emphasizes credibility—such as phrases like "trusted by thousands" or "certified quality"—can foster confidence and reduce the perceived risk associated with online purchases. On the other hand, for younger or more impulsive audiences, emotionally charged, reward-based language such as "We're giving away a special gift to the first 50 buyers!", can be more effective in prompting quick engagement and purchases. Streamers and manager could build a pre-evaluation system based on the research conclusions of this paper, taking full account of consumers' emotional susceptibility, that is, predicting user engagement and product sales by identifying the most persuasive linguistic combinations of product descriptions. Accordingly, live streaming e commerce platforms can formulate corresponding marketing strategies to help them obtain higher product sales.

This study provides an insight on the importance of employing exaggerated and emotional language, especially when discussing rewards. The notable positive influence of such language on audience engagement and product sales indicates that streamers should incorporate expressions that create a sense of urgency and excitement, such as "once-in-a-lifetime offer," "limited-time deal," or "exclusive chance." These expressions trigger emotional reactions and encourage purchasing behavior, making them essential for sustaining the lively nature of live-stream shopping. Based on the findings of this study, managers could establish foundational guidelines for live streaming and product descriptions, assisting streamers in presenting products effectively. Applying Aristotle's principles of rhetoric, streamers could elevate the persuasive impact of their speech. For instance, when describing rewards or products, they could use heightened and emotive language to announce discounts and prizes, which would evoke consumer enthusiasm. The findings show that combining credibility with an element of exaggeration positively influences sales, suggesting that streamers should strategically weave credible statements with subtle exaggeration to enhance product allure while maintaining trustworthiness. An example of this approach could be using phrases like "this is one of the most dependable product types I've found," blending reliability with excitement. Managers could provide training on effectively incorporating persuasive language, emphasizing how to use exaggeration strategically to enhance, rather than undermine, trust. This approach ensures that exaggeration is perceived as a credible element of the narrative rather than as a deceptive tactic.

Nonetheless, the study also warns streamers about overusing logical language in their product descriptions, as this can negatively affect engagement and sales. While conveying clear product details is important, streamers should steer clear of explanations that are too detailed or technical, as these can overwhelm or lose the interest of viewers. Simply put, audiences may not be drawn to logical justifications for rewards. Therefore, streamers are advised to avoid excessive explanations of reward mechanics during promotions, as this approach may weaken the emotional connection. Instead, streamers should aim for a balance between providing essential information and using emotionally engaging language to keep their audience interested. Additionally, streamers should prioritize brief, compelling statements that highlight key product benefits rather than lengthy and detailed descriptions. This approach helps ensure that viewers remain attentive while still receiving necessary product information.

Interactive methods play a crucial role in enhancing customer

engagement, which has been shown to mediate the connection between persuasive language combinations and product sales. Streamers can employ interactive tools such as Q&A sessions and personalized shoutouts to cultivate a more engaging environment. When these techniques are paired with persuasive language strategies that blend combinations like reward and exaggeration, reward and emotion, credibility and exaggeration, or credibility and emotion, they create a captivating experience. This immersive approach helps retain viewer attention and motivates them to make purchases.

The result highlights the significant role of electronic word of mouth (eWOM) as a moderating factor that enhances the impact of engagement on sales. Therefore, streamers should encourage viewers to leave ratings, comments, and reviews on product page and real-time bullet chat, which can then be highlighted during live streaming to strengthen credibility and build trust. This approach helps other viewers recognize that real consumer is sharing positive opinions, thereby reinforcing the streamer's credibility-based language with authentic peer validation. For mangers, ensuring the post-sales service quality of the live stream can improve its public praise rating which will help guide consumers from engagement to purchase behavior. For live streaming e commerce platform, implementing monitoring and feedback mechanisms can support continuous improvement in language use. Developing analytics tool to track how specific phrases and language combinations correlate with viewer engagement and sales spikes enables streamers to refine their language strategies over time based on actual performance data. This feedback loop allows streamers experiment with different approaches, adapt their communication style, and consistently improve their persuasive effectiveness. Strategically leveraging eWOM can reassure hesitant buyers and strengthen the persuasive impact of a streamer's language, leading to increased engagement and conversions.

6.4. Limitations and future research

Although this paper innovatively uses Aristotle's rhetorical theory to study the influence of persuasive language styles on customer engagement and product sales in the context of live streaming with goods, there are still shortcomings. First, this study was conducted in China, which may limit the generalizability of the findings. Cultural differences can significantly influence how audiences perceive and respond to different types of persuasive language, such as exaggeration, logic, or emotional appeals. As the judgment criteria of audiences may be dependent on cultural, ethnic, and gender differences. Future research should conduct cross-cultural comparisons to explore how language strategies impact consumer behavior in diverse cultural settings. This could include examining variations in responses between Western and Eastern audiences or analyzing linguistic effects in multilingual live-streaming markets.

Second, the data collection for this study occurred during the Double 11 Festival, a time known for its heightened consumer activity and promotional intensity. This context may have influenced the results, as consumer behavior during such large-scale events often differs from typical shopping periods. Future research should include data from regular live-streaming sessions and compare these with event-specific sessions to understand whether the effects of persuasive language vary between high-intensity promotional events and more routine, day-to-day live-stream broadcasts.

Third, this study focused exclusively on linguistic aspects of persuasion, without considering other multimodal factors such as the streamer's tone of voice, body language, pacing, and visual presentation. Future research could adopt a multimodal approach that integrates these factors to better understand how they interact with language strategies to affect consumer engagement and purchasing behavior. This would provide a more comprehensive view of persuasive language in live-streaming, capturing the nuanced ways in which visual and auditory cues combine with language to enhance the effectiveness of a streamer's performance.

Fourth, future research could explore longitudinal effects of persuasive language strategies on customer loyalty and repeat purchasing behavior. While this study focused on immediate engagement and sales, understanding the lasting impact of persuasive language on customer retention and brand loyalty would be beneficial for developing long-term marketing strategies. Researchers could also examine how changes in language use over time influence these outcomes.

Fifth, future studies could investigate the role of advanced technologies, such as AI-driven chatbots and virtual streamers, in replicating or enhancing human persuasive language strategies. As the use of AI in live-streaming grows, understanding how artificial agents can mimic or improve upon human-like persuasive communication could offer significant implications for the future of digital marketing and e-commerce.

CRediT authorship contribution statement

Xiangyang Ma: Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization. **Jiawen Ren:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Xianghui Khor:** Writing – review & editing, Formal analysis, Conceptualization. **Ruina Wang:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tieshan Li:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization. **Xiaoping Lang:** Writing – review & editing, Supervision, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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